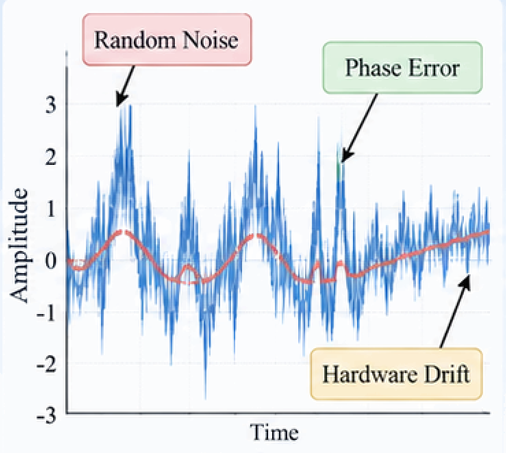
Preprocessing–Model Coupling in CSI Sensing

Raw CSI

Task-relevant human-motion variation mixed with noise, phase errors, and hardware drift.

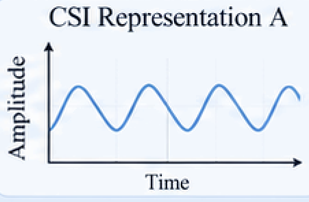


Preprocessing Pipelines

Different preprocessing pipelines generate distinct CSI representations.

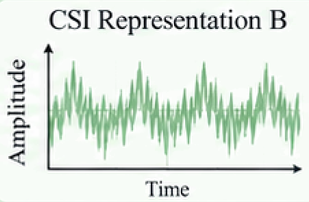
Pipeline A

Denoise → Filter



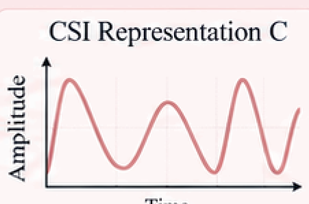
Pipeline B

Phase Correction → Offset Removal



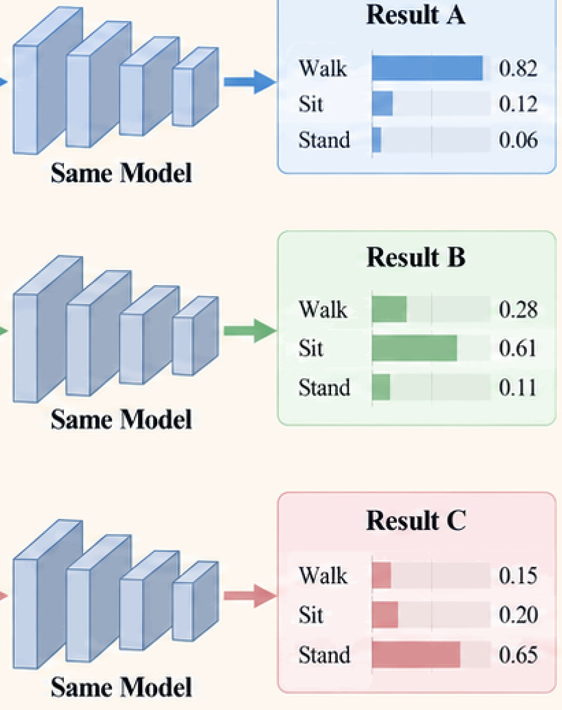
Pipeline C

Normalize → Resample



Controlled Recognition

Same Architecture · Same Data Split · Same Training



Model-Type Effect

Different model types have different recognition capability, further affecting the result differences.

